

Stress Test One: Mathematics in Every Shape

A. Tester

B. Checker

Abstract

We collect many kinds of mathematical notation in one place, such as $\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$, to see whether a reading tool keeps every symbol. The abstract itself contains a sum $\sum_{k=1}^n k = \frac{n(n+1)}{2}$.

1 Inline mathematics

Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be a smooth function and let $x_0 \in \mathbb{R}^n$. The gradient $\nabla f(x_0)$ points in the direction of steepest ascent, and the Hessian $H_{ij} = \partial^2 f / \partial x_i \partial x_j$ describes the curvature. For a vector $\mathbf{v}$ with $\|\mathbf{v}\|_2 = 1$ we have $D_{\mathbf{v}} f(x_0) = \langle \nabla f(x_0), \mathbf{v} \rangle$. Greek letters such as $\alpha, \beta, \gamma, \Delta, \Omega$ and ε appear often, and so do sets like $\mathcal{F}, \mathcal{L}, \mathfrak{g}$ and $\mathbb{C}$. Accents such as $\hat{x}, \tilde{y}, \bar{z}, \vec{u}$ and $\dot{q}$ must survive, as must primes $f'(x)$ and $g''(t)$. A small inline fraction $\frac{1}{2}$, a normal one $\frac{a+b}{c-d}$ and a limit $\lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n = e$ sit inside the text. Probability: $\Pr[X \geq t] \leq \frac{\mathbb{E}[X]}{t}$ for $t > 0$, and $X \sim \mathcal{N}(\mu, \sigma^2)$. Indices can be deep: $a_{i_1 i_2}^{(k)}, x^{2^n}, e^{-\frac{x^2}{2\sigma^2}}$ and $\sqrt[3]{x^3 + y^3}$.

2 Display equations

The Gaussian integral is

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}. \quad (1)$$

Equation (1) is classical. A double and a contour integral:

$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \oint_{\partial D} P dx + Q dy. \quad (2)$$

Without a number:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}, \quad \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}} = \zeta(s).$$

An aligned derivation with several numbers:

$$(a + b)^2 = a^2 + 2ab + b^2, \quad (3)$$

$$(a - b)^2 = a^2 - 2ab + b^2, \quad (4)$$

$$a^2 - b^2 = (a + b)(a - b). \quad (5)$$

A piecewise definition:

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{otherwise.} \end{cases} \quad (6)$$

Matrices and determinants:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}, \quad \det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc. \quad (7)$$

A continued fraction and a binomial sum:

$$x = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3}}}, \quad \sum_{k=0}^n \binom{n}{k} = 2^n. \quad (8)$$

Braces over and under terms:

$$\underbrace{1 + 1 + \cdots + 1}_{n \text{ times}} = n, \quad \overbrace{x \cdot x \cdots x}^m = x^m.$$

A long equation broken over lines:

$$p(x) = 3x^6 + 14x^5y + 590x^4y^2 + 19x^3y^3 - 12x^2y^4 - 12xy^5 + 2y^6 - a^3b^3 - a^2b^4 + ab^5. \quad (9)$$

3 Theorems

Theorem 3.1 (Cauchy–Schwarz). *For all vectors u, v in an inner product space, $|\langle u, v \rangle|^2 \leq \langle u, u \rangle \cdot \langle v, v \rangle$.*

Proof. If $v = 0$ both sides vanish. Otherwise put $\lambda = \langle u, v \rangle / \langle v, v \rangle$ and expand $0 \leq \|u - \lambda v\|^2$. $\square$

Lemma 3.2. *The function $F(t) = \int_0^t \frac{\sin s}{s} ds$ tends to $\frac{\pi}{2}$ as $t \rightarrow \infty$.¹*

4 Lists with formulas

1. The derivative of x^n is nx^{n-1} for every $n \in \mathbb{N}$.
2. Euler's identity $e^{i\pi} + 1 = 0$ links five constants.
3. The Fourier transform $\hat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx$.

The final sentence ends with a formula: the solution of $ax^2 + bx + c = 0$ is

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \quad (10)$$

¹This is the Dirichlet integral; note that $\int_0^\infty \left| \frac{\sin s}{s} \right| ds = \infty$.